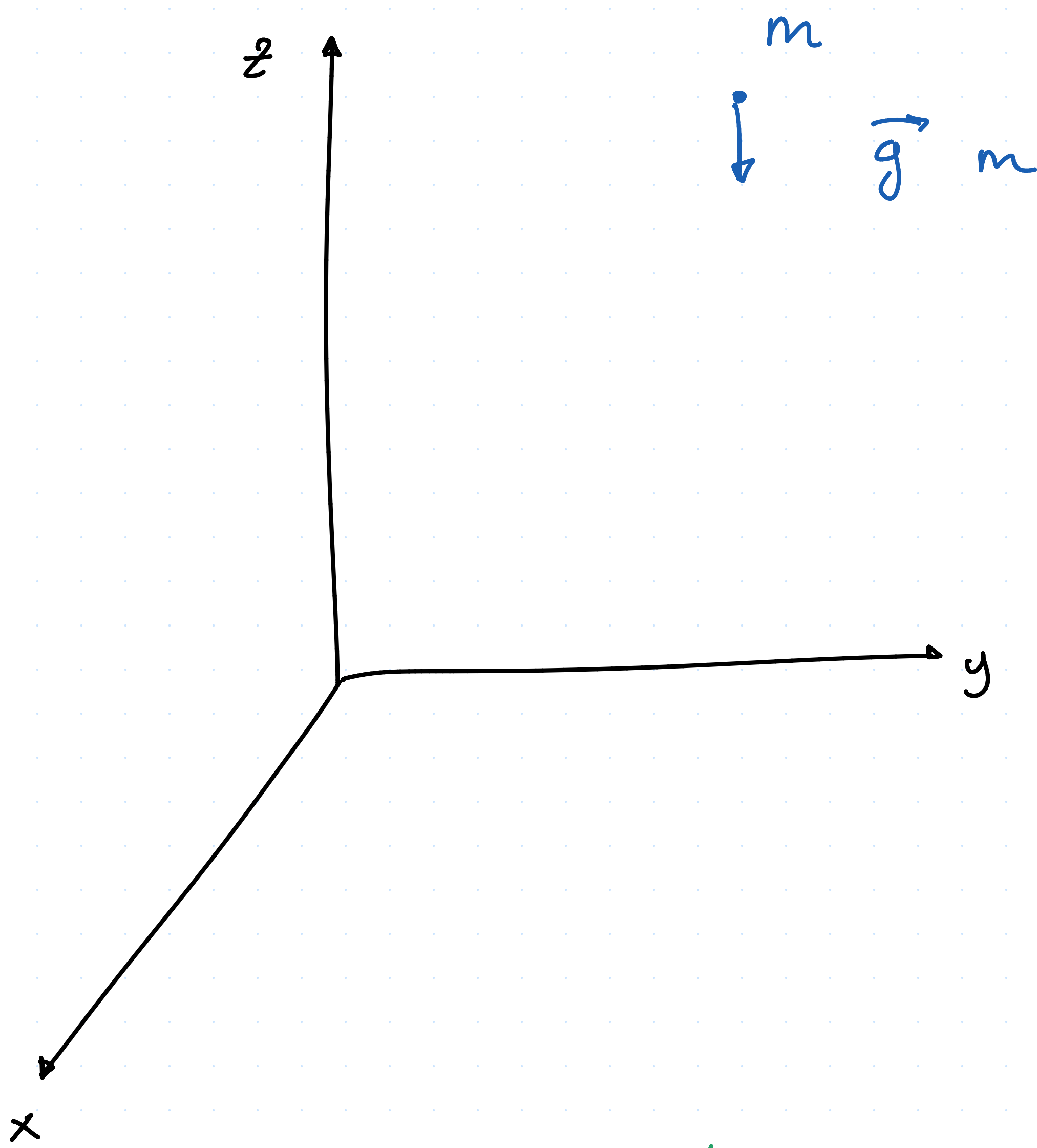


$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}_i} \right) - \frac{\partial \mathcal{L}}{\partial x_i} = 0 \quad \underline{i = 1, 2, 3}$$

$$\vec{r} = x \hat{x} + y \hat{y} + z \hat{z} = r \cos \phi \sin \theta \hat{x} + r \sin \phi \sin \theta \hat{y} + r \cos \theta \hat{z}$$



Cartesianas rectangulares

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_i} \right) + \frac{\partial U}{\partial x_i} = 0$$

$$L = T - U$$

Gravitación universal en esféricas

$$\vec{F}_m = -G \frac{mM\hat{r}}{r^2} = -G \frac{mM}{x^2+y^2+z^2} \left(\frac{x\hat{x} + y\hat{y} + z\hat{z}}{\sqrt{x^2+y^2+z^2}} \right)$$

Cartesianas rectangulares.

$$(x, y, z) \rightarrow (q_1, q_2, q_3)$$

$\parallel \quad \parallel \quad \parallel$
 $r \quad \theta \quad \phi$

mapa siempre y cuando el
Jacobiano sea distinto de
0.

$$x = r \cos \phi \sin \theta$$

$$y = r \sin \phi \sin \theta$$

$$z = r \cos \theta$$

$$J = r^2 \sin \theta \neq 0$$

$$x = x(q_1, q_2, q_3, t)$$

$$y = y(q_1, q_2, q_3, t)$$

$$z = z(q_1, q_2, q_3, t)$$

$$J \neq 0$$

Jacobiano $\neq 0 \Rightarrow \exists$ inversa y existe
difeomorfismo.

vector de posición

$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$\vec{r} = \vec{r}(x, y, z)$$

$$\vec{r} = \vec{r}(q_1, q_2, q_3, t) \} \text{ Campo vectorial.}$$

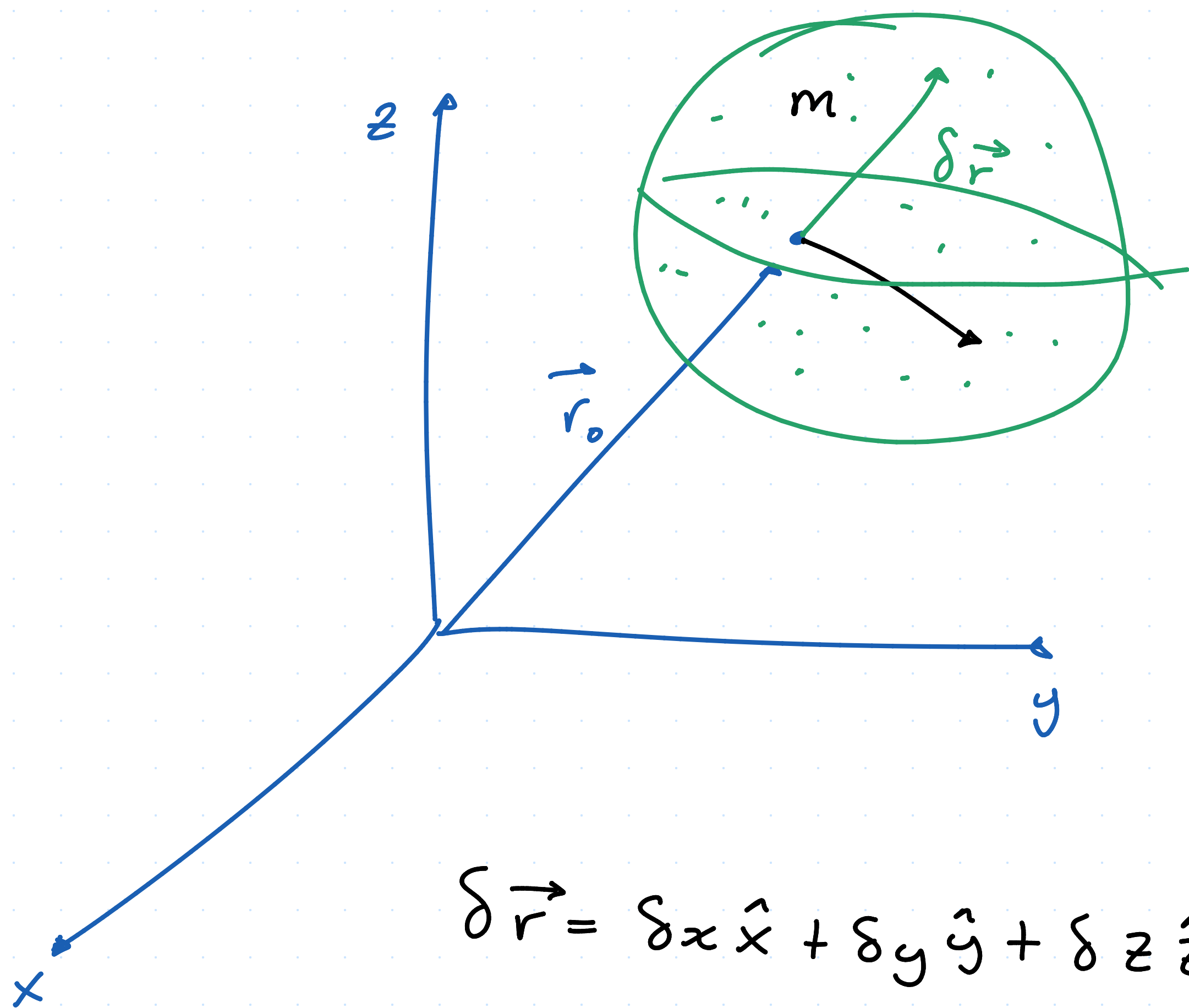
Velocidad por regla de la cadena

$$\dot{\vec{r}} = \frac{d\vec{r}}{dt} = \frac{\partial \vec{r}}{\partial q_1} \dot{q}_1 + \frac{\partial \vec{r}}{\partial q_2} \dot{q}_2 + \frac{\partial \vec{r}}{\partial q_3} \dot{q}_3 + \frac{\partial \vec{r}}{\partial t}$$

Ahora respecto a $\dot{q}_1$

$$\frac{\partial \dot{\vec{r}}}{\partial \dot{q}_1} = \frac{\partial \vec{r}}{\partial q_1} \left(\frac{\partial \dot{q}_1}{\partial \dot{q}_1} \right) = \frac{\partial \vec{r}}{\partial q_1}$$

• 3 grados de libertad.



$$\delta \vec{r} = \delta x \hat{x} + \delta y \hat{y} + \delta z \hat{z}$$

$$\hat{x} \cdot \delta x = \left(\frac{\partial x}{\partial q_1} \delta q_1 + \frac{\partial x}{\partial q_2} \delta q_2 + \frac{\partial x}{\partial q_3} \delta q_3 \right) \hat{x}$$

$$\hat{y} \cdot \delta y = \left(\frac{\partial y}{\partial q_1} \delta q_1 + \frac{\partial y}{\partial q_2} \delta q_2 + \frac{\partial y}{\partial q_3} \delta q_3 \right) \hat{y}$$

$$\hat{z} \cdot \delta z = \left(\frac{\partial z}{\partial q_1} \delta q_1 + \frac{\partial z}{\partial q_2} \delta q_2 + \frac{\partial z}{\partial q_3} \delta q_3 \right) \hat{z}$$

Desplazamiento
virtual

$$\Rightarrow \delta \vec{r} = \frac{\partial \vec{r}}{\partial q_1} \delta q_1 + \frac{\partial \vec{r}}{\partial q_2} \delta q_2 + \frac{\partial \vec{r}}{\partial q_3} \delta q_3$$

Tenemos:

$$m \frac{d^2 \vec{r}}{dt^2} = \vec{F}$$

Número
infinito de
desplazamientos

Trabajo virtual

segunda ley de Newton para
una partícula de masa m .

$$\left(m \frac{d^2 \vec{r}}{dt^2} \right) \cdot \delta \vec{r} = \underbrace{\vec{F} \cdot \delta \vec{r}}_{\delta w}$$

Volumen de una esfera usando esféricas vs cartesianas.

Describimos en términos de E. Cinética y Potencial.

$$\begin{aligned} \Rightarrow \delta w = \vec{F} \cdot \delta \vec{r} &\rightarrow \vec{F} \cdot \left[\frac{\partial \vec{r}}{\partial q_1} \delta q_1 + \frac{\partial \vec{r}}{\partial q_2} \delta q_2 + \frac{\partial \vec{r}}{\partial q_3} \delta q_3 \right] \\ &= \underbrace{\vec{F} \frac{\partial \vec{r}}{\partial q_1}}_{Q_1} + \underbrace{\vec{F} \frac{\partial \vec{r}}{\partial q_2}}_{Q_2} + \underbrace{\vec{F} \frac{\partial \vec{r}}{\partial q_3}}_{Q_3} \delta q_3 \end{aligned}$$

Coordenadas generalizadas

$$\Rightarrow Q_1 = \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_1}; \quad \Rightarrow Q_1 = \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_1} = -\nabla U(\vec{r}) \cdot \frac{\partial \vec{r}}{\partial q_1}$$

$$\vec{\nabla} U(\vec{r}) = \frac{\partial U}{\partial x} \frac{\partial x}{\partial q_1} + \frac{\partial U}{\partial y} \frac{\partial y}{\partial q_1} + \frac{\partial U}{\partial z} \frac{\partial z}{\partial q_1} = - \frac{\partial U}{\partial q_1}$$

$$\Rightarrow Q_i = - \frac{\partial U}{\partial q_i} ; \quad i = 1, 2, 3$$

Parte izquierda del trabajo virtual.

$$m \frac{d\vec{v}}{dt} \cdot \delta \vec{r} = m \frac{d\vec{v}}{dt} \cdot \left[\frac{\partial \vec{r}}{\partial q_1} \delta q_1 + \frac{\partial \vec{r}}{\partial q_2} \delta q_2 + \frac{\partial \vec{r}}{\partial q_3} \delta q_3 \right]$$

vel. vector de pos.

$$(*) \quad m \frac{d\vec{v}}{dt} \cdot \frac{\partial \vec{r}}{\partial q_1} = \frac{d}{dt} \left[m \vec{v} \cdot \frac{\partial \vec{r}}{\partial q_1} \right] = - m \vec{v} \cdot \frac{d}{dt} \left(\frac{\partial \vec{r}}{\partial q_1} \right)$$

$$\frac{\partial \dot{\vec{r}}}{\partial \dot{q}_1}$$

Por demostrar

$$\frac{\partial \dot{\vec{r}}}{\partial q_1}$$

Derivada total.

• sólo afecta al vector de pos.

$$\frac{\partial}{\partial \dot{q}_1} \left[\frac{1}{2} m \vec{v} \cdot \vec{v} \right] = \frac{1}{2} m \left[\frac{\partial \vec{v}}{\partial \dot{q}_1} \cdot \vec{v} + \vec{v} \cdot \frac{\partial \vec{v}}{\partial \dot{q}_1} \right]$$

$$= \frac{d}{dt} \left[\frac{\partial}{\partial \dot{q}_1} \left(\frac{1}{2} m \vec{v}^2 \right) \right] - \frac{\partial}{\partial q_1} \left(\frac{1}{2} m \vec{v}^2 \right)$$

$$\Rightarrow m \frac{d\dot{\vec{r}}}{dt} \cdot \frac{\partial \vec{r}}{\partial q_1} = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_1} \right) - \frac{\partial T}{\partial q_1}$$

$$\left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_1} \right) - \frac{\partial T}{\partial q_1} \right] \delta q_1 + \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_2} \right) - \frac{\partial T}{\partial q_2} \right] \delta q_2 + \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_3} \right) - \frac{\partial T}{\partial q_3} \right] \delta q_3 = 0$$

$$\left. \frac{\partial T}{\partial q_3} \right\} \delta q_3 = - \frac{\partial U}{\partial q_1} \delta q_1 - \frac{\partial U}{\partial q_2} \delta q_2 - \frac{\partial U}{\partial q_3} \delta q_3 = 0$$

$\delta q_i = \text{variaciones independientes}$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_1} \right) - \frac{\partial T}{\partial q_1} = - \frac{\partial U}{\partial q_1}$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_2} \right) - \frac{\partial T}{\partial q_2} = - \frac{\partial U}{\partial q_2}$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_3} \right) - \frac{\partial T}{\partial q_3} = - \frac{\partial U}{\partial q_3}$$

$$\mathcal{L} \equiv T - U = T(q_i, \dot{q}_i, t)$$

Suponemos que $U(q_i)$
 en E. potencial no depende
 de velocidad y tiempo

Llegamos a

$$\boxed{\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0}$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_1} \right) - \frac{\partial T}{\partial q_1} = Q_1 \equiv \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_1} = \frac{d}{dt} \left(\frac{\partial U}{\partial \dot{q}_1} \right) - \frac{\partial U}{\partial q_1}$$

También tenemos

¿Qué sucede con una carga en
 un campo EMV?

• Partícula acelerada emite radiación.

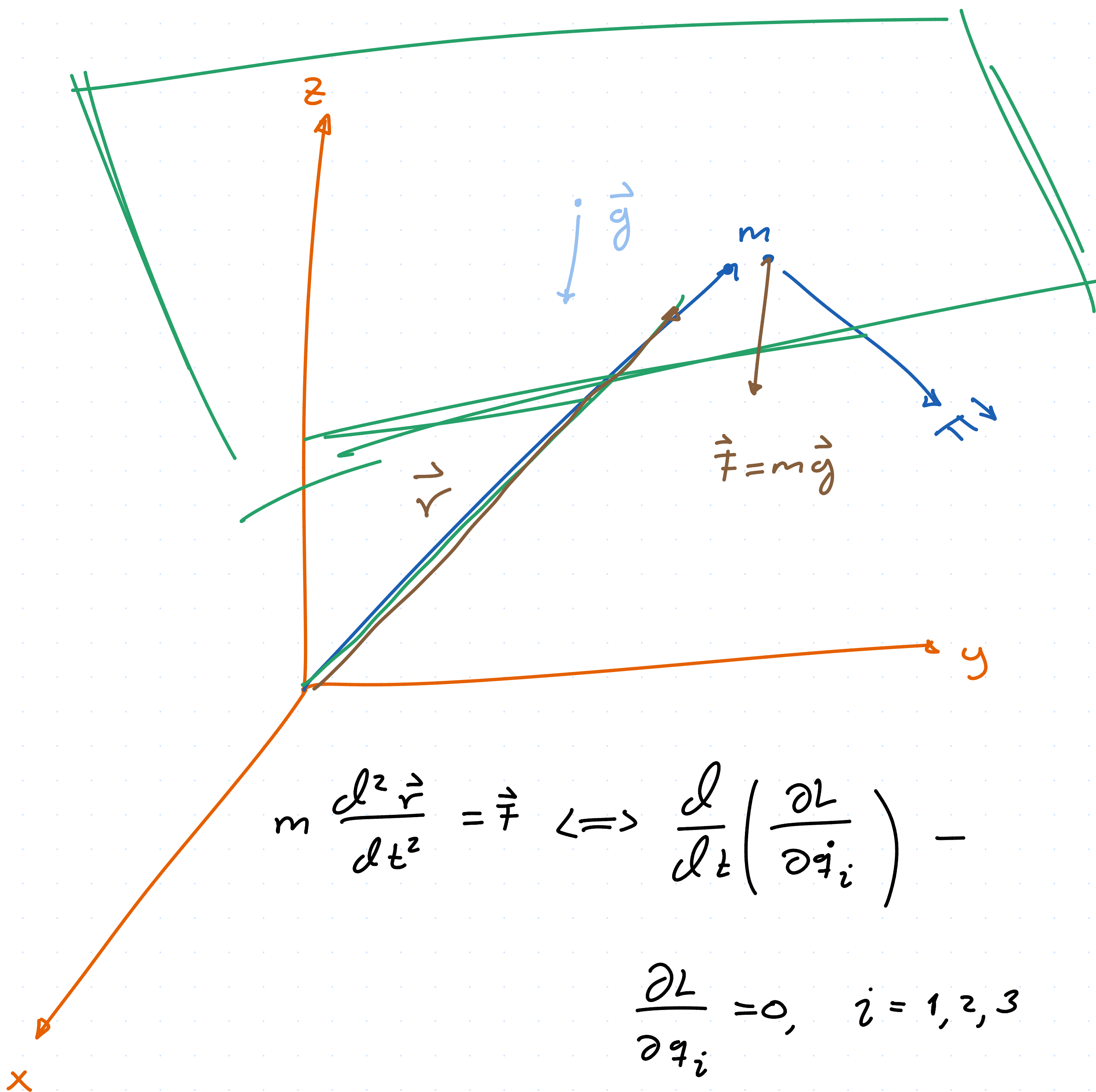
$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_2} \right) - \frac{\partial T}{\partial q_2} = Q_2 = \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_2} = \frac{d}{dt} \left(\frac{\partial U}{\partial \dot{q}_2} \right) - \frac{\partial U}{\partial q_2}$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_3} \right) - \frac{\partial T}{\partial q_3} = Q_3 = \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_3} = \frac{d}{dt} \left(\frac{\partial U}{\partial \dot{q}_3} \right) - \frac{\partial U}{\partial q_3}$$

Teoría del potencia (var. compleja)

$$L = T(q_i, \dot{q}_i, t) - U(q_i, \dot{q}_i, t)$$

Ejemplo



$$\vec{g} = - \left(9.8 \frac{m}{s^2} \right) \hat{z}$$

$$\Rightarrow L \equiv T - U$$

$\uparrow$
cinética

$\uparrow$
potencial

• Aprender bien la base vectorial

$$\dot{\vec{r}} = \dot{x} \hat{x} + \dot{y} \hat{y} + \dot{z} \hat{z}, \quad \vec{r} = x \hat{x} + y \hat{y} + z \hat{z}$$

cinética

$$T = \frac{1}{2} m \dot{\vec{r}}^2 = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2); \quad \vec{F} = m \vec{g} = - \nabla U$$

$$0 \hat{x} + 0 \hat{y} + mg(-\hat{z}) = - \frac{\partial U}{\partial x} \hat{x} - \frac{\partial U}{\partial y} \hat{y} - \frac{\partial U}{\partial z} \hat{z}$$

$$0 = - \frac{\partial U}{\partial x}; \quad 0 = - \frac{\partial U}{\partial y}; \quad -mg = - \frac{\partial U}{\partial z}$$

$$mg = \frac{\partial U}{\partial z} = \underbrace{\frac{dU}{dz}}_{\text{Derivada total.}}$$

$$\Rightarrow \int_{u_0}^u du = \int_{z_0}^z mg dz$$

$$u - u_0 = mg(z - z_0); \quad \underline{U = mgz}$$

Lagrangiana

$$T = \frac{1}{2} m \dot{\vec{r}}^2 = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

$$L = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

Tenemos por Euler-Lagrange...

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0 \quad ; \quad i = 1, 2, 3$$

$$q_1 = x, \quad q_2 = y, \quad q_3 = z$$

$$\left. \begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} &= 0 \\ \frac{\partial \mathcal{L}}{\partial x} &= 0 \end{aligned} \right\} \quad ; \quad \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} = p_x$$

momento lineal
en x.
↓

Esto equivale a

$$m\ddot{x} = 0$$

$$m\ddot{y} = 0$$

Tenemos

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{z}} \right) - \frac{\partial \mathcal{L}}{\partial z} = 0$$

$$\frac{d}{dt} (m\dot{z}) - (-mg) = 0$$

$$m\ddot{z} + mg = 0$$

Fuerza de Lorentz:

$$\vec{F} = q \left(\vec{E} + \frac{1}{c} \vec{v} \times \vec{B} \right)$$

$$(\vec{v}, \vec{r})$$

$$(\vec{E}, \vec{B})$$